Collision energy revisited

antocciphysics

UNIT 4 / DAY 17 / HONORS

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use energy as well as momentum to classify a collision.

SUCCESS CRITERIA - I can...

1. Calculate total momentum before and after.
2. Calculate total kinetic energy independently.
3. Account for reduced kinetic energy without losing total energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

With right positive, a spring has k=**150 N/m** and displacement x=**-0.05 m**. **Find signed spring force** and direction.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $p_{total,\,i} = p_{total,\,f}$; $K = \frac{1}{2}mv^2$.
Negligible external impulse conserves momentum. Only elastic collisions also conserve total kinetic energy.

Worked reasoning: Two **1 kg** carts at **+2** and **0 m/s** stick. Momentum predicts final speed **1 m/s**. Kinetic energy decreases from **2** to **1 J**; momentum can be conserved while kinetic energy decreases.

Your turn: Draw separate momentum arrows and kinetic-energy bars before and after.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Equal-mass carts approach at equal opposite speeds and stick. **What happens** to total momentum and kinetic energy?

Before contact: add velocity arrows.



Cart A Cart B

Marzano check - circle: ① ② ③

04 Apply the physics

A **2 kg** cart at **3 m/s** sticks to a **1 kg** cart at rest.
1. **Find v f**, K_{before}, K_{after} and energy transformed.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

1 / 2

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Continue down each column.

05 Explain your reasoning

Why can a collision be inelastic even when the objects bounce apart?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **4 kg** cart starts at rest. A **26 N** pull and **8 N** resistance act over **4 m**. **Find final speed**.
Show the energy changes and transfers; **label units**.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

A **1 kg** cart at **4 m/s** hits an identical stationary cart. Afterward velocities are **1** and **3 m/s**.
1. **Check momentum** and energy; classify.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **2 kg** cart at **4 m/s** sticks to a **1 kg** cart at **-2 m/s**.
1. **Find final velocity** and fraction of initial kinetic energy lost.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3